METRIC

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SUPERSEDING
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(See 6.4)

MILITARY SPECIFICATION

PARTS, HARD RUBBER,

FOR USE WITH SUBMARINE STORAGE BATTERIES (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers natural or synthetic hard rubber parts including covers, cover fittings, saddles, support pins, and ventilation ducts and fittings for use in submarine storage battery compartments.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Corrugated, Triple Wall, 350 Pound Maximum Weight.

MILITARY
MIL-P-116 - Preservation, Methods of
MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D639 - Battery Containers Made From Hard Rubber or Equivalent Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, G.F. Earl, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Classes and Rules.

(Application for copies should be addressed to the ATA Tariff Section, 1616 P Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Synthetic and natural hard rubber used in the covers and parts shall be free from metallic particles and all granular matter and from lumps or aggregations not thoroughly incorporated into the rubber compound during mixing. Reclaimed materials shall be used to the maximum extent possible.

3.2 Hard rubber parts shall be in conformance with the drawings specified (see 6.2.1).

3.3 Tolerances. Unless otherwise shown on the drawings, the tolerances for hard rubber parts shall be as follows:

Dimensions (millimeter)	Tolerance (millimeter)
10 or less	± 0.5
Greater than 10 but less than 50	± 1.0
50 or greater	± 2.0

3.4 Composition. Iron shall not exceed 0.3 percent. Manganese shall not exceed 0.003 percent when tested in accordance with 4.4.2.

3.5 Solubility. The total loss of mass shall not exceed 6.2 grams per square meter (g/m^2) when tested in accordance with 4.4.3.

3.6 Tensile strength. Tensile strength of all hard rubber parts shall be not less than 2.75×10^7 pascal (Pa) when tested in accordance with 4.4.4.

3.7 Elongation. Elongation of all hard rubber parts shall be not less than 2.5 percent when tested in accordance with 4.4.5

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3.8 Transverse deflection. Test pieces representing hard rubber parts, including covers and vent duct material, shall withstand a force of 620 newtons (N) without breaking, and shall show an average deflection of not less than 4 millimeters (mm) at 620 N force. This requirement is not applicable to electrolyte withdrawal tubes, small threaded plugs used in covers, or any of the component parts of airlift pump assemblies (inner tube, outer tube, coupling nuts, washer, and shipping plugs) when tested in accordance with 4.4.6.

3.9 Environmental requirements. Hard rubber covers or parts shall not be adversely affected when subjected to environmental temperatures of minus 30°C to 55°C (see 4.3.3).

3.10 Marking. Molded parts shall have the piece number or other identifying marking molded as raised or impressed characters on the surface. Parts manufactured from sheets and tubes shall be marked with the piece number in an ineradicable manner, as by stamping or branding. If marking would impair the usefulness of the part, nonpermanent marking, as by stencil or tag, may be used.

3.11 Workmanship. All surfaces shall present a neat, smooth, and finished appearance, and shall be free from injurious pitting, fissures, blowholes, rough spots, rubber scale, blisters, and other deformations which may be caused by the presence of air pockets in the compound and other air pockets which may be formed between the rubber and the mandrel.

3.11.1 The interiors of ventilation ducts shall be made against polished mandrels to give the rubber compound a smooth, lustrous surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Visual and dimensional examination. All molded parts made in a production run shall be considered a lot for purposes of visual and dimensional examination.

4.3.1.2 Tests. Hard rubber compound produced in one process batch and offered for delivery at one time shall be considered a lot for purposes of tests. The lot size shall be equal to the mass of the lot in kilograms (kg).

4.3.2 Sampling.

4.3.2.1 Visual and dimensional examination. Sampling and acceptance shall be in accordance with MIL-STD-105 with an acceptable quality level (AQL) of 1.5 percent defective for samples examined as specified in 4.4.1.

4.3.2.2 Tests. Sampling and acceptance shall be in accordance with MIL-STD-414 with an AQL of 1.5 percent defective. Inspection level I shall be used. The samples shall be tested as specified in 4.4.2 through 4.4.6. Hard rubber compound from which test pieces are prepared shall be taken after the mixture has been prepared and made ready to build into parts and shall be vulcanized with and under the same conditions as the rubber parts. A slab for testing purposes shall be supplied for each vulcanizer heat of covers, vent ducts, and parts. The size of this test slab shall be such as to produce twice the number of samples required by MIL-STD-414. At the discretion of the Government and when the size of the rubber part permits, test samples may be cut from the finished rubber parts. Test samples may also be made from rejected samples. The portion of test slab not consumed in producing test samples shall remain intact and be put in safe keeping for use with parts requiring reheat and for retesting purposes. In every case where it becomes necessary to reheat parts for the elimination of any defects or blemishes, the remainder of the test slab representing the parts to be reheated shall undergo the identical treatment as the parts.

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4.3.3 Environmental conditioning. Samples to be tested for tensile strength, elongation, and transverse deflection (see 4.4.4, 4.4.5 and 4.4.6) shall first be stored for 24 hours at minus 30°C, allowed to warm up to ambient temperature for a period of 24 hours then stored for 24 hours at 55°C. After cooling to ambient temperature for a period of 24 hours the samples shall be tested.

4.4 Examination and test methods.

4.4.1 Visual and dimensional examination. Each of the sample molded parts selected in accordance with 4.3.2.1 shall be visually and dimensionally examined to verify compliance with 3.1, 3.2, 3.9 and 3.10. If the number of defective molded parts in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 Composition. Determination of total iron and manganese shall be in accordance with ASTM D639.

4.4.3 Solubility. Determination of mass loss shall be in accordance with ASTM D639 when 60 percent by mass sulfuric acid and 45 percent by mass potassium hydroxide are used for each specimen.

4.4.4 Tensile strength. The test for tensile strength shall be in accordance with ASTM D639.

4.4.5 Elongation. Elongation shall be determined in accordance with ASTM D639.

4.4.6 Transverse deflection. The tests shall be made on the test pieces after immersion in water at a temperature of 18°C - 24°C for at least 1 hour. Immediately after this immersion the pieces to be tested shall be tested in accordance with method 11051 of FED-STD-601. The temperature during the test shall be maintained at 18°C - 24°C.

4.4.7 Retesting. When rubber parts and the representative test piece slabs (see 4.3.2.2) have been reheated for any purpose, test pieces shall be prepared from the slab and subjected to tensile strength and elongation and transverse deflection tests.

4.5 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A. Parts shall be cleaned, dried and unit protected as follows. Parts shall be individually packaged except when used in sets or quantities greater than 1, unless otherwise specified (see 6.2.1).

5.1.1.1 Cleaning and drying. The cleaning process and drying procedure shall be in accordance with MIL-P-116. Selections of the cleaning process and drying procedure shall be at the contractor's option.

5.1.1.2 Preservative. Preservative compounds shall not be used. Parts when unit protected in sets or quantities greater than 1 shall have the parts dusted with talcum or soapstone powder.

5.1.1.3 Unit protection. Parts shall be wrapped in minimum 90-pound basis weight waxed or plastic coated kraft paper. Wraps shall be secured by means of pressure-sensitive tape. Wrapped parts shall be cushioned and packaged in paperboard folding or set-up or fiberboard boxes at the contractor's option. Box closure shall be accomplished by means of pressure-sensitive tape. Unit protection, method of preservation-packaging, shall meet the requirements for method III of MIL-P-116.

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5.1.2 Level C cleaning and drying. Cleaning and drying shall be in accordance with 5.1.1.1.

5.1.2.1 Packaging. Packaging shall afford protection to the parts against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. Packaging may conform to the contractors normal retail procedure when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Parts packaged as specified (see 6.2.1) shall be packed in containers conforming to anyone of the following specifications at the option of the contractor:

Specification	Type or Class
PPP-B-585	Class 3 use
PPP-B-601	Overseas type
PPP-B-621	Class 2, overseas
PPP-B 636	Weather-resistant
PPP-B-640	Class 2

Wood constructed containers shall be provided waterproof caseliners in accordance with MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Fiberboard boxes shall be closed, waterproof and reinforced in accordance with the appendix to the applicable box specification except that method V shall apply for PPP-B-636 boxes. Options, where provided in the box specifications such as grades,styles, etc., shall be at the contractor's option. The gross weight of wood constructed boxes shall not exceed 90 kilograms (200 pounds); fiberboard boxes shall not exceed the gross weight limitations of the applicable fiberboard box specification.

5.2.2 Level B. Parts packaged as specified (see 6.2.1) shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 1
PPP-B-601	Domestic type
PPP-B-621	Class 1-domestic
PPP-B-636	Domestic
PPP-B-640	Class 1

Box closure shall be as specified in the applicable box specification or appendix thereto. Options, where provided in the box specification such as grades, styles, etc., shall be at the contractor's option. The gross weight of wood constructed boxes shall not exceed 90 kilograms (200 pounds); fiberboard boxes shall not exceed the gross weight limitations of the applicable fiberboard box specification.

5.2.3 Level C. Parts packaged as specified (see 6.2.1) shall be packed in containers acceptable to the common carrier which shall ensure safe delivery at the destination in a satisfactory condition at the lowest applicable rate. Containers or method of packing shall comply with Uniform Freight or National Motor Freight Classification rules or regulations or other carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Synthetic and natural rubber parts covered by this specification are intended for use in propulsion batteries, battery parts, battery repair or maintenance, and in battery compartments of submarines.

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6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Part required.
(c) Applicable drawings (see 3.2).
(d) Levels of preservation and packaging (see 5.1).
(e) Quantity per unit package if other than specified (see 5.1.1).
(f) Levels of packing (see 5.2).
(g) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
4.2	Inspection system plan	DI-R-4803	-----

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated the issue in effect of date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3,4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6140-N424)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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